

Mihai-Emilian Buduroi

github | *stackoverflow* | *linkedin*

be.mihai22@gmail.com | +4368120594406 | Based in **Vienna, Austria**

Embedded Systems Engineer with over five years of experience developing high-reliability firmware and low-level software for RTOS-based and bare-metal systems. Expertise spans embedded software architecture as well as hands-on hardware design and manufacturing, enabling end-to-end development from concept through production. Focused on hardware-software integration and delivering production-ready embedded systems.

Key Skills

- 5 years of embedded software engineering specializing in **C/C++** and **RTOS** development (FreeRTOS, Zephyr)
- Hands-on experience with **STM32**, Nordic **nRF52** series, **ESP32**, and **Atmel AVR** architectures, using a variety of embedded IDEs and toolchains
- Skilled in designing and debugging embedded communication interfaces (**I2C**, **SPI**, **UART**), using oscilloscopes and logic analyzers to catch signal issues, fix hardware-software mismatches, and ensure system reliability
- Certified in ***requirement engineering***
- Skilled in low-level embedded debugging using **JTAG/SWD** interfaces, **UART** serial tracing, **Segger J-Link** tools, and memory/core dump analysis
- Hands-on experience with **PCB design**, prototyping, and hardware validation, including board bring-up, initial firmware flashing, and functional testing
- Experienced with runtime debugging and memory analysis on Linux and Android systems using GDB and Valgrind
- Skilled in setting up **CI/CD** pipelines and containerized environments using GitHub/GitLab, Docker, and Jenkins to streamline embedded software delivery
- **Strong communicator**, skilled at simplifying complex technical concepts, leading discussions between engineering and business teams, and navigating conflicts to build consensus

Work Experience

Lead Software Engineer | **Aradon.bio** | **Leiden, The Netherlands** | **October 2024 – Present**

- Worked as the lead systems engineer for the backend platform, gathering technical requirements through stakeholder interviews, designing a scalable microservice-based architecture, and specifying APIs and communication protocols for device integration. Delivered production-ready systems used in lab automation.
- Designed, developed and secured GraphQL and REST APIs with token-based authentication and encryption. Deployed self-hosted, multi-region OpenVPN servers with automatic client provisioning and device isolation to ensure reliable device connectivity.
- Deployed a Dockerized backend architecture, enabling consistent production and testing environments.
- Automated deployment pipelines with GitHub Actions and built a two-command Docker setup, cutting developer onboarding and platform deployment time to under 25 minutes.
- Built AI-powered workflows to speed up backend architecture work, generate repeatable microservice code, and make integration testing faster — while keeping the codebase clean, reliable, and easy to maintain.

Embedded Systems Software Engineer | Kapsch TrafficCom AG | Vienna, Austria | December 2023 – December 2024

- Maintained and extended C++ firmware running on a FreeRTOS-based GNSS-enabled on-board unit (OBU) used in international truck tolling.
- Engineered real-time fuel and weight monitoring features, improving toll accuracy and supporting regulatory compliance across multiple European markets.
- Developed a fault-detection mechanism for charging ports that reduced false tolling events by 90% and customer complaints by over 30%.
- Delivered multilingual UI and firmware updates to achieve Swiss certification standards, decreasing field deployment issues by 70%.
- Built and maintained GitLab CI pipelines and Jenkins workflows for automated regression testing and secure OTA updates.
- Coordinated release management and product specifications within an 8-person team, working directly with external stakeholders and clients.

Lead R&D Engineer | WeMBED Systems S.R.L | Austria, Germany, The Netherlands | July 2020 – December 2023

- Founded and led an embedded systems company, applying self-taught electronics design skills alongside formal Computer Science studies.
- **Research:** Explored embedded systems technologies, working with MCUs from Infineon, NXP, Nordic and Texas Instruments; developed three patent applications for Panorama Laboratories BV, including:
 - “Add-on device for attachment on a micropipette/suction device” (WO2020067900A3)
 - “Micropipette or add-on device for a micropipette comprising at least two magnetometers” (WO2024030029A1)
 - “Bench layout determination” (WO2024030030A1)
- **Development:** Led the end-to-end development of five market-ready embedded products, covering hardware design, firmware engineering, and production delivery, generating €130K in annual revenue.
- **Production & Manufacturing:** Set up an in-house production lab for small-to-mid volume manufacturing, and built early access relationships with hardware vendors at Embedded World and Electronica.
- **Operations:** Built a secure AWS-based server infrastructure and deployed self-hosted OpenVPN servers to protect device communications and internal data transfers.
- **Leadership & Business:** Led a cross-functional team of four across business and technical roles; handled accounting, customer acquisition, and strategic partnerships

DevOps Engineer & Backend Developer | magic42 LTD | Birmingham, UK | October 2018 – July 2019

- Supported and improved a PHP/Laravel backend system, delivering stability and new feature updates for a 30-member organization.
- Built and optimized Jenkins-based CI/CD pipelines on AWS, streamlining build processes and reducing deployment errors.
- Refactored the production database for GDPR compliance: conducted two security audits and remediating four user-data vulnerabilities.
- Mentored a junior developer, providing technical guidance and accelerating their integration into the project team.
- Coordinated release schedules with external stakeholders to minimize downtime and ensure smooth deployments.

IT Project Manager (Fixed-Term Contract) | Study Abroad & Exchanges, University of Birmingham | Birmingham, UK | February 2018 – June 2018

- Deployed a centralized, ERASMUS-compliant database to replace manual Excel tracking.
- Coached staff on SQL scripting for data automation, significantly reducing manual errors.
- Optimized program website SEO to boost organic search visibility.

Education

BSc Computer Science with a Year in Industry at University of Birmingham - 2016 - 2020

- Major Project: **Project Hawk** - Researched, designed and integrated a drone for studying cell tower suitability for commercial drone applications. **Dissertation**
- Graduated with First Class Honours, focusing on Kernel Programming, Network and Security, Embedded Software Engineering, and Computer Systems Architecture.
- Besides low level architecture and languages of computer design, the degree exposed me to key concepts and hands-on experience for helping teams and businesses. Knowing how to implement Agile and Waterfall methods in organizations helps me deliver for big budget projects, as well as for fast and nimble startups

A Level equivalent (Romanian Baccalaureate / High School Level) in Computer Science and Advanced Maths - 2012 - 2016

- 91% in C programming
- 90% in Advanced Maths
- C1 English Language proficiency through IELTS (2017)

Self-taught skills

Technical Growth

- CAD modelling and advanced FDM 3D printing as well as basic CNC manufacturing techniques
- Self-taught PCB design and electronics development, gaining hands-on experience through access to a hardware penetration lab.

Personal Growth

- Strengthened communication and conflict resolution skills through client-facing work, focusing on active listening and minimizing workplace stress during critical situations.
- Improved productivity by restructuring work pipelines, prioritizing tasks effectively, and applying lean management principles inspired by The Goal by Eliyahu M. Goldratt.

Languages

- *English* (Fluent), *French* (Intermediate), *Romanian* (Native), *German* (Conversational)

Hobbies

- Endurance sports (triathlon), psychology and philosophy books, cooking and diverse gastronomies